Week 12

MICROECONOMIC THEORY
ECON 323 502/503

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Review

- Representing a game with a payoff matrix.
- A **dominant strategy** is a strategy that is optimal no matter what an opponent does.

Prisoners' Dilemma

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -20
	Deny	-20, 0	-1, -1

Prisoner 1 has a dominant strategy -- Confess

Prisoner 2 has a dominant strategy – Confess

A natural prediction of the game is that both players play their dominant strategies – which is an equilibrium in dominant strategies.

Equilibrium in Dominant Strategies

- **Equilibrium in dominant strategies** is the outcome of a game in which each player plays a dominant strategy. → This is a natural prediction of a game.

- A player may not always have a dominant strategy. (For example, the Rock-Paper-Scissors game.)

- Not every game has an equilibrium in dominant strategies. (Again, the Rock-Paper-Scissors game is such an example.)

Prisoners' Dilemma

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -20
	Deny	-20, 0	-1, -1

Prisoners' Dilemma has an equilibrium in dominant strategies (Confess, Confess).

Is an equilibrium in dominant strategies (Confess, Confess) a good outcome for the players? To answer this, we need to define what “good” is.

Pareto Domination

- An outcome of a game, x, **Pareto dominates** another outcome, y, if x makes at least one player better off without hurting any other player compared to y.

- In this case, we also say y **is Pareto dominated by** x.

- E.g. An outcome with payoff profile (11, 20) Pareto dominates (10, 19)
- E.g. An outcome with payoff profile (10, 20) Pareto dominates (10, 19)
- E.g. An outcome with payoff profile (1, 2) does not Pareto dominate (2, 1)

Prisoners' Dilemma

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -20
	Deny	-20, 0	-1, -1

In prisoners' dilemma, we have four feasible outcomes (we do not consider randomization).

- Is (Confess, Confess) Pareto dominated by another outcome?
  It is Pareto dominated by (Deny, Deny).
- Is (Confess, Deny) Pareto dominated by another outcome? No
- Is (Deny, Confess) Pareto dominated by another outcome? No
- Is (Deny, Deny) Pareto dominated by another outcome? No

Prisoners' Dilemma

		Prisoner 2	
		Confess	Deny
Prisoner 1	Confess	-5, -5	0, -20
	Deny	-20, 0	-1, -1

Two important features of Prisoners' Dilemma:

1. It has an equilibrium in dominant strategies;
2. The equilibrium in dominant strategies is Pareto dominated by another feasible outcome.

Practice Problem 1: Does the following game satisfy both features of a prisoners' dilemma?

		Husband	
		Clean	Don't Clean
Wife	Clean	10, 10	0, 15
	Don't Clean	15, 0	2, 2

A. Yes
B. No

Practice Problem 1: Does the following game satisfy both features of a prisoners' dilemma?

		Husband	
		Clean	Don't Clean
Wife	Clean	10, 10	0, 15
	Don't Clean	15, 0	2, 2

A. Yes
B. No

Yes, both features are satisfied:
(1) Wife has a dominant strategy (Don't Clean). Husband has a dominant strategy (Don't Clean).
(2) The equilibrium in dominant strategies (Don't Clean, Don't Clean) is Pareto dominated by another outcome (Clean, Clean).

Practice Problem 2: Does the following game satisfy both features of a prisoners' dilemma?

		Husband	
		Clean	Don't Clean
Wife	Clean	10, 10	0, 15
	Don't Clean	15, 0	12, 12

A. Yes
B. No

Practice Problem 2: Does the following game satisfy both features of a prisoners' dilemma?

		Husband	
		Clean	Don't Clean
Wife	Clean	10, 10	0, 15
	Don't Clean	15, 0	12, 12

A. Yes
B. No

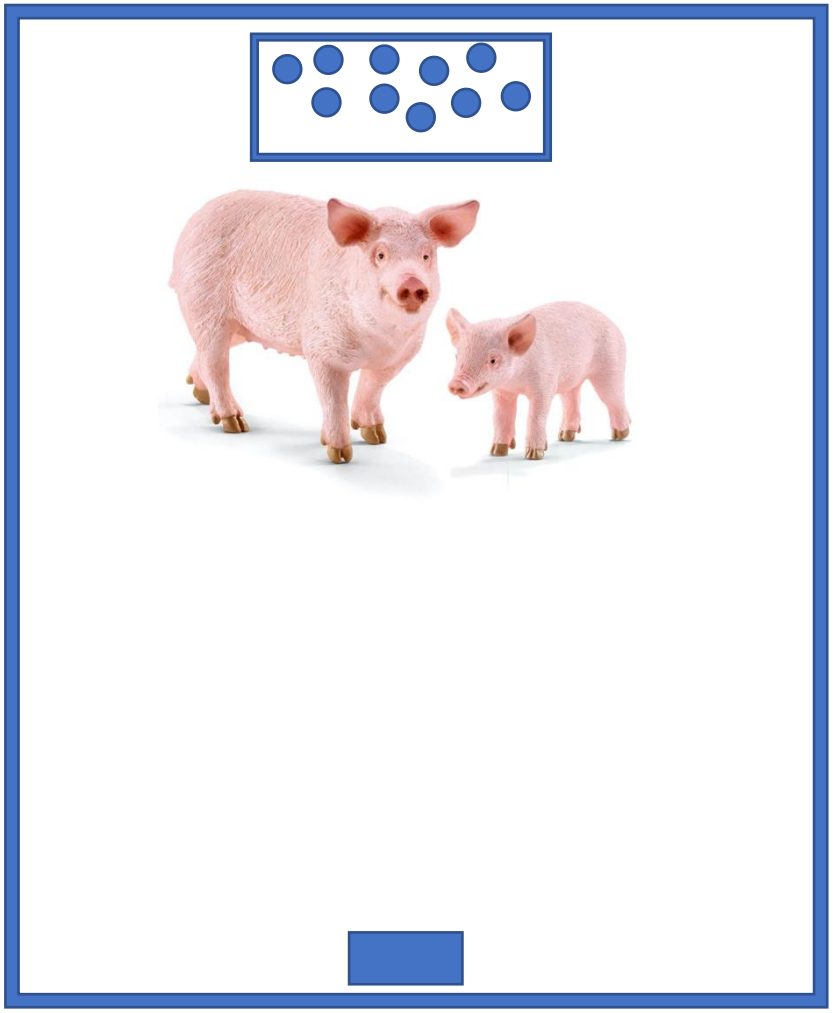
No, because only one feature is satisfied:
(1) Wife has a dominant strategy (Don't Clean). Husband has a dominant strategy (Don't Clean).
(2) The equilibrium in dominant strategies is not Pareto dominated by another feasible outcome.

Equilibrium in dominant strategies is a natural prediction of the game, but such an equilibrium may not always exist.

However, nonexistence of an equilibrium in dominant strategies does not mean that we can say nothing about what will happen in this game.

The Pig and Piglet Game

Dispenser



Panel

		Piglet	
		Wait	Press
Pig	Wait	0, 0	10, -1
	Press	1, 8	6, 2

1. Does the Pig have a dominant strategy?
Answer: No. The Pig's best strategy depends on the strategy of the Piglet.

2. Does the Piglet have a dominant strategy?
Answer: Yes. It is optimal for the Piglet to wait regardless of the Pig's strategy.

3. Is there an equilibrium in dominant strategies?
Answer: No. Because the Pig has no dominant strategy.

4. Can you predict what will happen?
One potential answer: As the Pig anticipates the Piglet to play its dominant strategy (wait), it is optimal for the Pig to press the panel. (Press, Wait) will show up.

		Piglet	
		Wait	Press
Pig	Wait	0, 0	10, -1
	Press	1, 8	6, 2

How did pigs play the game in actual animal experiments?

The Pig did most of the pressings!!!

https://www.sciencedirect.com/science/article/pii/0003347279900332

		Piglet	
		Wait	Press
Pig	Wait	0, 0	10, -1
	Press	1, 8	6, 2

We will introduce a weaker prediction of a game: **Nash equilibrium**.

As a preparation:

A **best response** is the best strategy to take given the other player's strategy.

If the Piglet chooses to **Wait**, then the best response of the Pig is to **Press**.
If the Piglet chooses to **Press**, then the best response of the Pig is to **Wait**.
If the Pig chooses to **Wait**, then the best response of the Piglet is to **Wait**.
If the Pig chooses to **Press**, then the best response of the Piglet is to **Wait**.

Nash Equilibrium

A **Nash Equilibrium** is a profile of strategies in which each player is playing a **best response** to the other player's strategy.

(Named after Mathematician John Nash, a Nobel Laureate, Biographical Movie: A Beautiful Mind)

In the two-player case, we say (S_1, S_2) is a Nash equilibrium, if

(1) given Player 2 plays S_2, it is a best response for Player 1 to play S_1,

(2) given Player 1 plays S_1, it is a best response for Player 2 to play S_2.

		Piglet	
		Wait	Press
Pig	Wait	0, 0	10, -1
	Press	1, 8	6, 2

Let us verify if (Press, Wait) is a Nash equilibrium.

(1) Given the Piglet chooses to **Wait**, the best response of the Pig is to **Press**.

(2) Given the Pig chooses to **Press**, the best response of the Piglet is to **Wait**.

Hence, both players are playing their best response to the other player's strategy. (Press, Wait) is a Nash equilibrium.

		Piglet	
		Wait	Press
Pig	Wait	0, 0	10, -1
	Press	1, 8	6, 2

Is (Wait, Wait) is a Nash equilibrium?
No. Given the Piglet chooses to Wait, the Pig's best response should be to **Press**.

Is (Press, Press) is a Nash equilibrium?
No. Given the Pig chooses to Press, the Piglet's best response should be to **Wait**.

Is (Wait, Press) is a Nash equilibrium?
No. Given the Pig chooses to Wait, the Piglet's best response should be to **Wait**.